

Hyper-V Virtual Network Configuration

After successfully installing the Hyper-V role, virtual switches were created to support Internet connectivity and network segmentation for the enterprise lab environment.

1.External Virtual Switch Configuration

An External Virtual Switch was created and bridged to the host machine's physical network adapter to provide Internet access to virtual machines.

Virtual Switch Details

Switch Name	Type	Purpose
WAN-SWITCH	External	Internet Connectivity

Configuration Procedure

1. Open **Hyper-V Manager**.
2. Select **Virtual Switch Manager**.
3. Choose **External** and click **Create Virtual Switch**.
4. Enter the switch name:

WAN-SWITCH

5. Select the physical network adapter of the host machine.
6. Enable **Allow management operating system to share this network adapter**.
7. Click **Apply** and then **OK**.

Result

The WAN-SWITCH was successfully created and connected to the physical network adapter, providing Internet access for the virtual infrastructure.

2. Internal Virtual Switch Configuration

Five Internal Virtual Switches were created to provide isolated network segments for servers and departments.

Virtual Switch Details

Switch Name	Type	Purpose
SERVER-SWITCH	Internal	Server Network
MAMT-SWITCH	Internal	Management Network

Switch Name	Type	Purpose
FIN-SWITCH	Internal	Finance Network
HR-SWITCH	Internal	Human Resources Network
IT-SWITCH	Internal	Information Technology Network

Configuration Procedure

1. Open **Hyper-V Manager**.
2. Select **Virtual Switch Manager**.
3. Choose **Internal** and click **Create Virtual Switch**.
4. Enter the switch name.
5. Click **Apply** and then **OK**.
6. Repeat the process for all required internal switches.

Result

The following internal switches were successfully created:

- SERVER-SWITCH
- MAMT-SWITCH
- FIN-SWITCH
- HR-SWITCH
- IT-SWITCH

These switches provide isolated network segments for different departments within the enterprise lab environment.

